

Ribociclib (Kisqali) – Drug Summary

Ribociclib is a selective small-molecule inhibitor of cyclin-dependent kinases 4 and 6 (CDK4/6) used in **hormone receptor-positive, HER2-negative** breast cancer. It is FDA-approved in combination with endocrine therapy: (1) **adjuvant** therapy for adult patients with HR⁺/HER2⁻ stage II–III early breast cancer at high risk of recurrence ¹, and (2) **advanced/metastatic** therapy (HR⁺/HER2⁻) with an aromatase inhibitor or fulvestrant ² ³. In both settings, ribociclib is added to endocrine agents (e.g. letrozole or fulvestrant) to enhance efficacy. It was originally approved (2017) for postmenopausal women (with LHRH agonists for premenopausal women and men) and recently expanded to the adjuvant high-risk HR⁺ population. Ribociclib's oral administration and clinical trials (MONALEESA-2/3/7, NATALEE) have shown improved progression-free and overall survival in luminal (ER⁺/PgR⁺) subtypes ⁴. There is **no current indication** for HER2-positive or triple-negative (basal-like) breast cancers; in fact, TNBC often features RB1 loss or CDK6 upregulation conferring intrinsic resistance to CDK4/6 inhibitors (see **Resistance** below). Mechanistically, ribociclib blocks the cyclin D–CDK4/6–Rb axis, causing **G1 arrest** and reduced proliferation ⁵.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL3545110 ⁶.
- **DrugBank ID:** DB11730 ⁷.
- **Brand names:** Kisqali (Novartis) ⁸ (sold in 200 mg tablets; e.g. Kisqali 200 mg daily dose; also co-packaged with letrozole as the Kisqali Femara co-pack ⁸).
- **Generic name:** Ribociclib (LEE-011).
- **Synonyms/Development codes:** LEE-011; LEE-011A; LEE011; LEE011A ⁹.

Mechanism of Action (Breast Cancer Context)

Ribociclib is a **selective inhibitor of CDK4 and CDK6**, kinases that bind D-type cyclins and drive G₁→S cell-cycle progression ⁵. In ER⁺ breast cancer, estrogen signaling upregulates cyclin D1, increasing CDK4/6 activity. By inhibiting CDK4/6, ribociclib **prevents phosphorylation of the retinoblastoma protein (Rb)**, causing Rb to remain in its hypophosphorylated, E2F-bound state ⁵. This blocks E2F-mediated transcription and G₁/S transition, leading to cell-cycle arrest. In vitro studies show ribociclib *decreases* pRb levels and proliferation in breast cancer cells ¹⁰. Clinically, adding ribociclib to endocrine therapy has yielded deeper and more durable cell-cycle arrest (e.g. greater Ki-67 suppression) and longer progression-free survival in HR⁺/HER2⁻ disease ⁴. Notably, ribociclib also causes concentration-dependent QT prolongation (see Safety), and can have off-target effects on other kinases at high doses, but its primary antitumor effect is through CDK4/6 blockade ⁵.

Primary Targets (Homo sapiens)

- **CDK4** (*Cyclin-dependent kinase 4*; HGNC: CDK4) – Key G₁/S regulator. Ribociclib binds the ATP pocket of CDK4, inhibiting its kinase activity ⁵.

- **CDK6** (*Cyclin-dependent kinase 6*; HGNC: CDK6) – Closely related kinase partnering with D-cyclins. Ribociclib similarly inhibits CDK6 ⁵.

Both CDK4 and CDK6 share ~71% identity and function redundantly in phosphorylating Rb ¹¹. No other primary targets have been validated for ribociclib in humans, though high CDK6 expression can confer resistance (see Resistance).

Pathways (Overview)

Ribociclib's effects intersect several cellular pathways. Key pathways include:

- **Cell cycle (KEGG: hsa04110; GO:0007049)** – The G₁/S transition pathway is directly inhibited. Ribociclib downregulates gene sets in the cell-cycle pathway (e.g. DNA replication, S-phase genes) via CDK4/6 blockade ⁵.
- **RB/E2F pathway (HALLMARK_E2F_TARGETS; Reactome R-HSA-69205)** – With CDK4/6 inhibited, Rb remains active, repressing E2F. Thus, E2F-target genes are suppressed ⁵.
- **G2/M checkpoint (HALLMARK_G2M_CHECKPOINT)** – Downstream of G₁, fewer cells enter S/G₂M under ribociclib, reducing expression of mitosis regulators (e.g. cyclin B, PLK1) in sensitive cells ⁵.
- **PI3K/AKT/mTOR signaling (HALLMARK_PI3K_AKT_MTOR_SIGNALING; KEGG: hsa04151)** – A major growth-survival axis. Activation of PI3K/AKT/mTOR can bypass CDK4/6 inhibition, and is a known resistance mechanism ¹².
- **FGFR/MAPK signaling (Reactome FGFR1 pathway; HALLMARK_KRAS_SIGNALING)** – FGFR activation (via FGF or amplification) stimulates downstream MAPK/ERK and PI3K signaling. Aberrant FGFR/MAPK activity is implicated in ribociclib resistance ¹³.
- **Estrogen response (HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE)** – ER α -driven transcription is the context for CDK4/6i use. While ribociclib does not directly inhibit ER pathways, it is used in ER⁺ (luminal) breast cancers to augment endocrine blockade.
- **Apoptosis (HALLMARK_APOPTOSIS)** – While ribociclib's primary action is cytostatic, it can indirectly promote apoptosis in some contexts. (No specific citation; apoptotic gene signatures may rise following prolonged cell-cycle arrest.)
- **Immune and inflammation (HALLMARK_INTERFERON_GAMMA_RESPONSE; REACTOME_INNATE_IMMUNE_SYSTEM)** – Genes involved in antigen presentation, interferon signaling, and innate immunity have been found to be **downregulated** by ribociclib in some studies ¹⁴ (see Downregulated section).
- **Antigen presentation (Reactome R-HSA-1236972; GO:0019882)** – Ribociclib treatment was associated with decreased expression of antigen-presenting cell (APC) gene sets ¹⁴.
- **DNA damage response/p53 pathway (GO:0006974; HALLMARK_P53_PATHWAY)** – Disruption of the cell cycle may engage DNA damage checkpoints; p53–p21 regulation influences CDK4/6i sensitivity (see p21 below ¹⁵).

Each of these pathways will be detailed below in terms of ribociclib-induced changes, and their roles in sensitivity or resistance.

Upregulated Pathways (Drug Effect)

While ribociclib primarily *inhibits* proliferation pathways, certain stress-response and inhibitory pathways can be upregulated in treated cells:

- **Cell-cycle inhibitors:** In sensitive tumors, CDK inhibitor pathways (e.g. **CDKN1A/p21** expression) may increase, reinforcing G₁ arrest. Indeed, ribociclib sensitivity correlates with higher p21 levels ¹⁵.
- **Senescence/apoptosis pathways:** Prolonged G₁ arrest can lead to cellular senescence and activation of apoptotic gene expression. Some preclinical data suggest increased expression of pro-apoptotic factors (e.g. Bax, cleaved caspases) in CDK4/6-treated cells, although specific ribociclib-induced signatures are not well catalogued.
- **Cyclin D1 accumulation:** A known feedback effect of CDK4/6 inhibition is nuclear accumulation of Cyclin D1 (unphosphorylated). Elevated Cyclin D1 does not overcome the block, but may prime cells to cycle if drug is withdrawn. (No direct citation available.)

(Note: Detailed transcriptomic data on ribociclib-upregulated genes are limited. Most studies emphasize downregulated proliferation and immune genes. The above are inferred from general cell-cycle biology and limited reports ¹⁵.)

Downregulated Pathways (Drug Effect)

Ribociclib consistently **downregulates cell proliferation pathways:**

- **Cell cycle progression:** G₁/S and G₂/M gene sets (e.g. Cyclins A/E, CDK2, DNA replication machinery) are markedly suppressed ⁵. Clinically, Ki-67 labeling index and PAM50 proliferation scores drop in ribociclib-treated tumors ¹⁶.
- **E2F targets:** Genes driven by E2F (required for S-phase entry) are downregulated via the Rb checkpoint ⁵.
- **Mitotic regulators:** Spindle and chromosome segregation pathways (e.g. PLK1, AURKA, BUB1) decrease as cells fail to enter S-phase.
- **Innate/Antigen-presenting immunity:** Gene sets for antigen-presenting cells (dendritic cells, macrophages) and **innate immune signaling** were found to decrease following ribociclib+letrozole in luminal-B tumors ¹⁴. Specifically, interferon-response and APC signature expression were *lower* after ribociclib, indicating immune suppression in the tumor microenvironment ¹⁴.
- **Inflammation/Interferon:** Interferon-alpha and -gamma pathways (Hallmark) were downregulated in correlation with complete cell-cycle arrest ¹⁴. (By contrast, these pathways rose post-chemotherapy, but fell with ribociclib.)
- **(Other):** No data were found indicating ribociclib downregulates DNA repair or metabolic pathways specifically. As ribociclib is cytostatic, many housekeeping pathways remain unchanged.

Citations: All downregulation statements are supported by mechanistic understanding ⁵ and recent clinical molecular analyses ¹⁶ ¹⁴.

Sensitivity Markers and Pathways

Pathways and markers **associated with increased ribociclib sensitivity** (typically upregulated or intact in responders) include:

- **High CDK4/Low CDK6 expression:** Cells expressing *high* levels of CDK4 but *low* CDK6 were found to be extremely sensitive to ribociclib ¹⁷. In fact, CDK6 overexpression conferred resistance in models ¹⁸, so the inverse (low CDK6) is a sensitivity marker.
- **p21 (CDKN1A) signaling:** Elevated p21 levels favor G₁ arrest. Clinical and preclinical studies show ribociclib-sensitive breast cancers have higher p21 expression, whereas resistant cells often lose p21 ¹⁵.

Thus, *upregulation of p21 or intact p53–p21 checkpoint* correlates with sensitivity ¹⁵ .

- **Estrogen receptor signaling:** Strong ER signaling (e.g. intact ESR1 without ligand-binding mutations) means cells remain hormone-driven and responsive to combined endocrine+CDK4/6 blockade. Although not a “pathway up/down”, HR⁺ status itself is the primary biomarker for CDK4/6i efficacy ⁴ .

- **Cyclin D1 amplification/overexpression:** Many HR⁺ tumors overexpress cyclin D1; these are dependent on CDK4/6 activity. In such tumors, CDK4/6 inhibition yields pronounced arrest. (Cyclin D1 overexpression per se is not required for approval, but it is common in responders ¹⁹ .)

Conversely, pathways whose **downregulation** is associated with sensitivity:

- **Cyclin E–CDK2 pathway:** Low activity of the Cyclin E–CDK2 axis favors sensitivity. Clinically, tumors *without* CCNE1 (cyclin E1) amplification respond better. In contrast, upregulation of cyclin E/CDK2 undercuts the need for cyclin D–CDK4/6 and predicts resistance ²⁰ .

- **PI3K/AKT signaling:** Tumors with low PI3K/AKT pathway activity (or those without PIK3CA/PTEN alterations) may rely more on the cyclin D–CDK4/6 pathway and thus be more sensitive. Inhibiting PI3K can restore sensitivity in resistant models. Downregulation of PI3K/AKT (e.g. via co-treatment) enhances ribociclib response ¹² .

References: Correlations of ribociclib sensitivity with high CDK4/low CDK6 and with p21 are documented ¹⁷ ¹⁵ . PALOMA-3 and MONALEESA trial biomarker analyses also support that low CCNE1 (cyclin E) favors benefit ²¹ .

Resistance Markers and Pathways

Acquired or intrinsic resistance to ribociclib is linked to upregulation or activation of alternative cell-cycle or survival pathways:

- **RB1 loss/mutation:** The most definitive resistance mechanism is loss of functional Rb. Without Rb, CDK4/6 inhibitors cannot enforce G₁ arrest. Clinical cases of acquired palbociclib/ribociclib resistance often harbor RB1 mutations ²² . (No pathway ID, but the Rb–E2F pathway is inactivated.)
- **CDK6 amplification/overexpression:** As noted, forced CDK6 expression drove resistance ¹⁸ . Thus, tumors that upregulate CDK6 can bypass CDK4 blockade (CDK6–cyclin D still gets inhibited by ribociclib, but overexpression may shift stoichiometry).
- **Cyclin E/CDK2 upregulation:** Amplification or overactivity of Cyclin E1 (CCNE1) and CDK2 drives Rb phosphorylation independently of CDK4/6. This is a well-known resistance pathway ²¹ (e.g. PALOMA-3 showed high CCNE1 correlates with shorter PFS).
- **p16 (CDKN2A) upregulation or RB pathway disruption:** Paradoxically, overexpression of p16 (INK4A) can lead to resistance if Rb is also lost ²³ . More broadly, any disruption of the Rb checkpoint (CDKN2A amplification, 12q13-15 deletion) confers insensitivity.
- **PI3K/AKT/mTOR activation:** Aberrant PI3K pathway signaling (e.g. PIK3CA mutations, PTEN loss) is enriched in CDK4/6 inhibitor-resistant tumors ¹² . Upstream growth factor activation (e.g. FGFR1 amplification) turns on PI3K/AKT and MAPK/ERK, providing alternative proliferation signals ¹³ .
- **FGFR/MAPK activation:** Overactive FGFR or HER family signaling can phosphorylate Rb via MAPK-driven cyclins, also causing resistance ¹³ .
- **p21 downregulation:** Loss of p21 (often via p53 mutation or other means) was observed in resistant models ²⁴ , allowing cells to progress through G₁ despite CDK4/6 inhibition.

Pathways whose **downregulation** contributes to resistance:

- **p53/p21 pathway:** Reduced activity of the p53–p21 checkpoint (e.g. TP53 mutation) undermines G₁ arrest and can promote CDK4/6i resistance ¹⁵.
- **ER signaling loss:** In some resistant metastatic lesions, ER expression falls; these become less driven by cyclin D and thus less sensitive to CDK4/6i.

Sources: Preclinical screens and clinical biomarker analyses have identified all the above mechanisms ²³ ²¹ ¹² ¹⁸ ¹⁵. For example, a study showed that breast cancer clones with low p21 and high CDK6 levels acquired resistance to long-term ribociclib, consistent with the above pathways ²⁴ ¹⁸.

Breast Cancer Subtype–Specific Evidence

- **Luminal A (HR⁺/HER2⁻, low proliferation):** These tumors generally respond well to ribociclib + endocrine therapy, often achieving durable disease control. Markers of high CDK4/cyclin D1 and intact Rb are common.
- **Luminal B (HR⁺/HER2⁻, higher proliferation/Ki67):** Such tumors are more proliferative and actually derive *greater* absolute benefit from CDK4/6 inhibition, as shown by larger Ki-67 drop and iDFS gains ¹⁶. In the CORALLEN neoadjuvant trial (luminal B tumors), ribociclib+letrozole markedly suppressed proliferation compared to chemotherapy ¹⁶, although some tumors showed reactivation of cell-cycle by surgery ¹⁶. Immune gene signatures in luminal B were specifically dampened by ribociclib ¹⁴.
- **HER2-positive (HER2⁺/HR⁻):** Ribociclib is not approved for HER2⁺ disease. Early-phase trials combining CDK4/6 inhibitors with anti-HER2 therapy (in HR⁺/HER2⁺) suggest some activity, but effects in HER2⁺/HR⁻ tumors are unproven. Many HER2⁺ tumors have concurrent cyclin D1 amplification but can signal through other kinases. (No formal indication.)
- **Triple-negative (HR⁻/HER2⁻):** TNBC generally *does not* benefit from CDK4/6 inhibitors. Most TNBC lack hormone receptors, often have Rb loss or low Rb activity, and rely on CDK2/E2F or other pathways. Indeed, preclinical data show minimal response of basal-like models to ribociclib. CDK4/6i are not recommended for TNBC outside trials.
- **BRCA-mutated or HRD-positive:** Germline BRCA1/2 mutant cancers are typically TNBC; in HR⁺ breast cancer, BRCA mutations are rare. There is no indication that BRCA status predicts for ribociclib efficacy. In fact, BRCA-mutant tumors often have defective Rb pathways. PARP inhibitors, not CDK4/6i, are the targeted therapy in BRCA-driven BC.

Contraindications and Safety

- **Contraindications:** *None listed* in FDA labeling ²⁵ ²⁶. Ribociclib has no absolute contraindications.
- **Warnings/Precautions:** Major risks include **myelosuppression**, **QT interval prolongation**, **hepatotoxicity**, and **ILD/pneumonitis**. Frequent Grade 3/4 adverse effects are neutropenia and leukopenia ²⁷. Liver enzymes (ALT/AST, bilirubin) can rise – LFTs must be monitored every 2 weeks for the first two cycles and periodically thereafter ²⁸. Ribociclib prolongs QTc: ECG monitoring at baseline and Day 14 of cycle 1 is recommended ²⁹. Concomitant use with other QT-prolonging drugs is discouraged (tamoxifen is contraindicated due to additive QT risk ³⁰). Cases of **severe cutaneous reactions** (SJS/TEN/DRESS) and **pneumonitis/ILD** have occurred; these require immediate discontinuation ³¹. Embryo-fetal toxicity is a concern – women of childbearing potential must use contraception, and breastfeeding is contraindicated ³² ³³.

- **Dose modifications:** Cytopenias usually recover after temporary interruption or dose reduction. Hepatic impairment requires dose reduction (recommended 400 mg/day for moderate/severe impairment ³⁴). Severe renal impairment (CrCl <30 mL/min) also calls for a reduced starting dose (200 mg/day) ³⁵.
- **Drug interactions:** Ribociclib is a **CYP3A4 substrate and moderate inhibitor**. Strong CYP3A4 inhibitors (e.g. ketoconazole) should be avoided or used with reduced ribociclib dose ³⁶. Strong CYP3A inducers should likewise be avoided. It also can increase exposure of other CYP3A substrates.

(All safety data are drawn from the FDA-approved label ³¹ ²⁹ and NCCN guidelines, which note careful monitoring for QTc, neutropenia, and LFT elevations.)

Trial and Guideline Context

- **Key Clinical Trials:** The MONALEESA program underpins ribociclib's use in metastatic HR⁺ breast cancer. MONALEESA-2 (NCT01958021) showed ribociclib+letrozole significantly prolongs PFS and OS vs letrozole alone in postmenopausal metastatic patients ³. MONALEESA-3 (NCT02422615) found ribociclib+fulvestrant improves outcomes in men and women with advanced HR⁺ disease ³. MONALEESA-7 (NCT02278120) demonstrated benefit in premenopausal women (with ovarian suppression). The NATALEE trial (NCT03701334) led to FDA's adjuvant approval: ribociclib+letrozole significantly improved 3-year invasive disease-free survival in high-risk early BC ³⁷. (Recent data show sustained disease-free benefit at 5 years.)
- **Regulatory:** FDA approved ribociclib for advanced BC in 2017 (Kisqali) and for adjuvant high-risk early BC in Sept 2024 ¹ ³⁷. EMA and PMDA approvals align with FDA. CHMP (EU) gave positive opinion for adjuvant use in late 2024.
- **Guidelines:** NCCN and ESMO breast cancer guidelines recommend CDK4/6 inhibitors (ribociclib, palbociclib, abemaciclib) as first-line therapy for HR⁺/HER2⁻ metastatic disease (Category 1) and as adjuvant therapy in high-risk early disease. (NCCN lists ribociclib + AI for node-positive or high-risk node-negative luminal BC, per Oct 2024 update.) ESMO's luminal BC guidelines similarly endorse CDK4/6 inhibitors with endocrine therapy. NICE (UK) has issued a technology appraisal (TA1086, Aug 2025) recommending adjuvant ribociclib + NSAI for high-risk early HR⁺/HER2⁻ BC ³⁸.
- **Emerging notes:** Ongoing trials are exploring ribociclib combinations (with PI3K/mTOR inhibitors, immunotherapy, etc.) and use in other settings (e.g. neoadjuvant luminal BC). Off-label, ribociclib was tested in HR⁺/HER2⁺ neoadjuvant regimens (NA-PHER2 trial), showing some Ki-67 suppression. No benefit has been seen in triple-negative or non-breast cancers (except a recent FDA approval in combination with letrozole for premenopausal ovarian cancer with HR⁺).

Additional Mechanistic/Clinical Notes

- **Biomarkers:** Apart from HR status, no clinical biomarker is validated for selecting ribociclib beyond endocrine sensitivity. Research suggests high RB1 expression, low cyclin E, high cyclin D1, and intact p21 may predict benefit. No FDA-approved predictive assay exists yet.
- **Overlap with other CDK4/6 inhibitors:** Ribociclib has a similar profile to palbociclib and abemaciclib, but differs pharmacologically (e.g. more QTc risk than palbociclib; abemaciclib causes more GI toxicity). Resistance patterns largely overlap, so cross-resistance is common, though some patients who progress on one may respond to another in rare cases.

- **Drug duration:** Ribociclib is given daily (3-weeks-on, 1-week-off cycle) until disease progression or unacceptable toxicity. Early evidence suggests continuous inhibition is needed, as “rebound” proliferation can occur during breaks ¹⁶.

Pathway Evidence Table

Pathway (ID)	Ribociclib Effect	Correlates with	Biological Rationale	References
Cell Cycle (KEGG hsa04110)	Down	Sensitivity (S)	Direct target of ribociclib. Inhibition of CDK4/6 arrests G1, lowering cell-cycle gene expression ⁵ .	32
E2F Targets (HALLMARK)	Down	Sensitivity (S)	CDK4/6 inhibition keeps Rb active, preventing E2F release; thus E2F-driven transcription is suppressed ⁵ .	32
G₂/M Checkpoint (HALLMARK)	Down	Sensitivity (S)	Fewer cells enter S/M phases under CDK4/6 blockade, so downstream mitotic genes are reduced ⁵ .	32
Interferon γ Response (HALLMARK)	Down	?	Ribociclib-treated luminal tumors show decreased IFN γ -related gene expression ¹⁴ ; may reflect immune suppression (unclear effect on therapy).	44
Innate Immune System (Reactome)	Down	Resistance (R)	Tumors on ribociclib exhibit reduced antigen-presenting cell (APC) and innate immune signatures ¹⁴ , potentially enabling immune escape.	44
PI3K-AKT-mTOR (HALLMARK)	Up	Resistance (R)	Upregulation of PI3K/AKT signaling provides alternative survival/proliferation signals, bypassing CDK4/6 arrest ¹² .	66
FGFR/MAPK Signaling (Reactome/HALLMARK_KRAS)	Up	Resistance (R)	FGFR amplifications/activation drive MAPK and PI3K pathways, reducing dependence on CDK4/6 ¹³ .	66

Pathway (ID)	Ribociclib Effect	Correlates with	Biological Rationale	References
Cyclin E/CDK2 (E2F Targets) (HALLMARK)	Up	Resistance (R)	CCNE1 (Cyclin E1) amplification activates CDK2–Rb phosphorylation, bypassing CDK4/6; high Cyclin E1 is linked to poor CDK4/6i response ²¹ .	48
p21 (CDKN1A) (HALLMARK_P53)	Up	Sensitivity (S)	Ribociclib sensitivity is proportional to p21 levels ¹⁵ ; p21 enforces G1 arrest. Tumors losing p21 become resistant.	71
p16 (CDKN2A) (INK4 Pathway)	Up	Resistance (R)	CDKN2A (p16) amplification along with Rb loss inactivates the CDK4/6–Rb axis, conferring CDK4/6i resistance ²³ .	46

Each entry identifies the pathway (with an example database ID), the net effect of ribociclib on that pathway (Upregulated or Downregulated), whether this change is associated with **Sensitivity** or **Resistance** to the drug, and the biological rationale with citations. “Up” means pathway activity increases in resistant or sensitive states; “Down” means ribociclib suppresses that pathway. For example, the cell-cycle pathway is **Down** by ribociclib and this is fundamental to the drug’s efficacy (Sensitive) ⁵, whereas PI3K–AKT–mTOR is **Up** in resistant tumors ¹².

Sources: FDA/EMA labels ^{1 31}, DrugBank ⁵, clinical trial reports ^{16 37}, and recent reviews of CDK4/6 inhibitors ^{4 23 21 12 17 15}. All human-specific data and HGNC gene symbols are used.

1 2 25 26 27 28 29 30 31 32 33 36 label

https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/209092s018lbl.pdf

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<https://www.onclive.com/view/fda-approves-adjuvant-ribociclib-in-hr-her2-breast-cancer>

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<https://www.nice.org.uk/guidance/ta1086>